

6.	Ans: C ACW moment due to $T = 20 \times 10^{-3}(9.81) \times 60$ $= 11.772 \text{ N cm}$ CW moment due to $W = 100 \times 10^{-3}(9.81) \times 10$ $= 9.81 \text{ N cm}$ At equilibrium, ACW = CW $11.772 = 9.81 + 50 \times 10^{-3}(9.81) \times d$ $d = 4 \text{ cm from the pivot}$ Hence, for the 50 g suspended mass to achieve a CW moment, it must be placed on the 44 cm mark on the rule.
7.	Ans: C